VISVESVARAYA TECHNOLOGICAL UNIVERSITY

"JnanaSangama", Belgaum -590014, Karnataka.

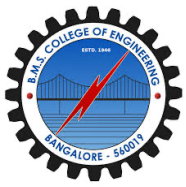
LAB REPORT
on

Object Oriented Java programing(23CS3PCOOJ)

Submitted by

Varsha M Gowda(1WA23CS033)

In partial fulfillment for the award of the degree of
BACHELOR OF ENGINEERING
in
COMPUTER SCIENCE AND ENGINEERING



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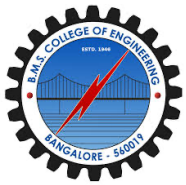

CERTIFICATE

This is to certify that the Lab work entitled "Object Oriented Java Programming (23CSPCOOJ)" carried out by **Varsha M Gowda(1WA23CS033),** who is bonafide student of **B.M.S. College of Engineering**. It is in partial fulfillment for the award of **Bachelor of Engineering in Computer Science and Engineering** of the Visvesvaraya Technological University, Belgaum. The Lab report has been approved as it satisfies the academic requirements in respect of a Database Management Systems(23CS3PCDBM) work prescribed for the said degree.

Lab faculty in charge Name Assistant Professor Department of CSE, BMSCE	Dr. Joythi S Nayak Professor & HOD Department of CSE, BMSCE

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Experiment 1

```
import java.io.*;
import java.util.Scanner;

class Quad{
	public static void main(String args[]){
		double a,b,c,s1,s2;
		Scanner sc = new Scanner(System.in);
		System.out.println("enter a b c of quadratic eqtn");
		a = sc.nextDouble();
		b = sc.nextDouble();
		c = sc.nextDouble();
		if(a==0){
			System.out.print("INVALID EQTN");
			return;
		}
		double d= b*b-4*a*c;
		if (d>0){
			System.out.println("real and distinct roots");
			s1=(-b + Math.sqrt(d))/(2*a);
			s2=(-b - Math.sqrt(d))/(2*a);
			System.out.print("roots are "+s1+" "+s2);
		}else if(d==0){
			System.out.println("real and equal roots");
			s1= -b/(2*a);
			System.out.print("root: "+s1);
		}else{
			s1= -b/(2*a);
			s2= Math.sqrt(-d)/(2*a);
			System.out.println("imaginary roots");
			System.out.println("root 1: "+s1+" +i "+s2);
			System.out.println("root 1: "+s1+" -i "+s2);
		}
	}
}
```

OUTPUT:

```
Enter values of a, b, and c:
1 2 1
The equation has equal roots: x1 = -1.0 x2= -1.0
```

```
Enter values of a, b, and c:
2 3 5
The equation has imaginary roots. There is no real solution.
```

```
Enter values of a, b, and c:

The equation has two real solutions: x1 = -1.0, x2 = -2.0
```

Experiment 2

```
import java.io.*;
import java.util.Scanner;

class    StudentD{
String usn, name;
int[] m1=new int[10],m2=new int[10];
int[] c1 = new int[10], c2 = new int[10];
int sub;

void accept(String n,String u, int s){

this.usn=u;
this.name=n;
this.sub=s;
Scanner sc = new Scanner(System.in);
System.out.println("enter details of sem1");
for(int i=0;i<s;i++){
        System.out.print("Enter marks for subject " + (i + 1) + ": ");
        m1[i] = sc.nextInt();
        System.out.print("Enter credits for subject " + (i + 1) + ": ");
        c1[i] = sc.nextInt();
}
System.out.println("Enter details for Semester 2:");
for (int i = 0; i < sub; i++) {
        System.out.print("Enter marks for subject " + (i + 1) + ": ");
        m2[i] = sc.nextInt();
        System.out.print("Enter credits for subject " + (i + 1) + ": ");
        c2[i] = sc.nextInt();
}

}

void display() {

System.out.println("USN: " + usn);
System.out.println("Name: " + name);

}

double sgpa(int sem){

double totalm=0,totalc=0;
if (sem==1){
for(int i=0;i<sub;i++){
totalm+=m1[i]*c1[i];
totalc+=c1[i];
}
```

```
}
else if (sem == 2) {
for (int i = 0; i < sub; i++) {
totalm += m2[i] * c2[i];
totalc += c2[i];
}
}
if (totalc == 0) return 0;
return totalm / (totalc*10);

}


double calcCgpa() {
double sgpaSem1 = sgpa(1);
double sgpaSem2 = sgpa(2);
return (sgpaSem1 + sgpaSem2) / 2;
}
}

public class Student {
public static void main(String[] args) {
Scanner sc = new Scanner(System.in);
System.out.print("Enter the number of subjects: ");
int sub = sc.nextInt();

Student student = new Student();
student.accept("Thanu","1WA23CS019", sub);
student.display();
double sgpaSem1 = student.sgpa(1);
double sgpaSem2 = student.sgpa(2);
System.out.println("SGPA for Semester 1: " + sgpaSem1);
System.out.println("SGPA for Semester 2: " + sgpaSem2);
double cgpa = student.calcCgpa();
System.out.println("CGPA: " + cgpa);
sc.close();
}
}
```

OUTPUT:

```
Enter USN: 1BM23CS003
Enter Name: Alice
Enter number of subjects: 5
Enter credits for subject 1: 4
Enter marks for subject 1: 78
Enter credits for subject 2: 4
Enter marks for subject 2: 88
Enter credits for subject 3: 3
Enter marks for subject 3: 66
Enter credits for subject 4: 3
Enter marks for subject 4: 45
Enter credits for subject 5: 2
Enter marks for subject 5: 100
Student Details:
USN: 1BM23CS003
Name: Alice
Subject-wise Details:
Subject 1: Credits = 4, Marks = 78
Subject 2: Credits = 4, Marks = 88
Subject 3: Credits = 3, Marks = 66
Subject 4: Credits = 3, Marks = 45
Subject 5: Credits = 2, Marks = 100
SGPA: 7.75
```

Experiment 3

```
import java.util.Scanner;
class Book {
   String name;
   String author;
   double price;
   int pages;
public Book(String name, String author, double price, int pages) {
      this.name = name;
      this.author = author;
      this.price = price;
      this.pages = pages;
   }
public String toString(){
return "name="+name+"\n"+"author="+author+"\n"+"price="+price+"\n"+"pages="+pages;
}
}
public class BookInfo {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);
      System.out.print("Enter the number of books: ");
      int n = sc.nextInt();
      sc.nextLine();
      Book[] books = new Book[n];
      for (int i = 0; i < n; i++) {
         System.out.println("enter details for book " + (i + 1) + ":");
         System.out.print("enter the name of book: ");
         String name = sc.nextLine();
         System.out.print("enter the name of author: ");
         String author = sc.nextLine();
         System.out.print("enter the price: ");
         double price = sc.nextDouble();
         sc.nextLine();
         System.out.print("enter the number of pages: ");
         int pages = sc.nextInt();
         sc.nextLine();
         books[i] = new Book(name, author, price, pages);
      }

      for (int i = 0; i < n; i++) {
         System.out.println("Details of book " + (i + 1) + ":");
         System.out.println(books[i].toString());

      }

      sc.close();
   }
}
```

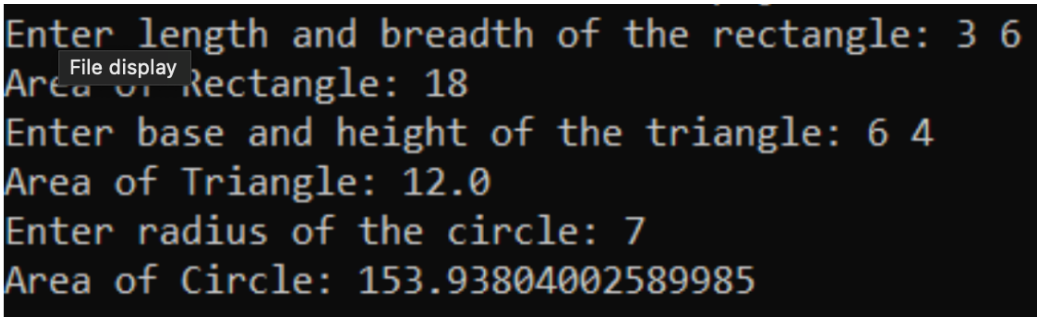
OUTPUT:

```
Enter number of books:
3
Book 1:
Enter name of Book: Brave New World
Enter Author name: Aldous Huxley
Enter price: 500
Enter number of pages: 250
Book 2:
Enter name of Book: Merchant of Venice
Enter Author name: Shakespeare
Enter price: 750
Enter number of pages: 400
Book 3:
Enter name of Book: The Da Vinci Code
Enter Author name: Dan Brown
Enter price: 999
Enter number of pages: 689
Book details:
Book name: Brave New World
Author name: Aldous Huxley
Price: 500.0
Number of pages: 250
Book name: Merchant of Venice
Author name: Shakespeare
Price: 750.0
Number of pages: 400
Book name: The Da Vinci Code
Author name: Dan Brown
Price: 999.0
Number of pages: 689
```

Experiment 4

```
import java.util.*;
abstract class Figure{
double dim1;
double dim2;
Figure(double a,double b)
{
dim1=a;
dim2=b;
}
abstract double area();
}
class rectangle extends Figure{
rectangle(double a,double b){super(a,b);}
double area()
{
System.out.println("inside area of rectangle:");
return dim1*dim2;
}
}
class triangle extends Figure{
triangle(double a,double b){super(a,b);}
double area()
{
System.out.println("inside area of triangle:");
return (dim1*dim2)/2;
}
}
class circle extends Figure{
circle(double a){super(a,a);}
double area()
{
System.out.println("inside area of circle:");
return Math.PI*(dim1*dim1);
}
}
class AbstractArea{
public static void main(String [] args){
rectangle r=new rectangle(10,20);
triangle t=new triangle(3,5);
circle c=new circle(9);
Figure figref;
figref=r;
System.out.println("area is:"+figref.area());
figref=t;
System.out.println("area is:"+figref.area());
figref=c;
System.out.println("area is:"+figref.area());
```

}}
OUTPUT:


Enter length and breadth of the rectangle: 3 6
Area of Rectangle: 18
Enter base and height of the triangle: 6 4
Area of Triangle: 12.0
Enter radius of the circle: 7
Area of Circle: 153.93804002589985

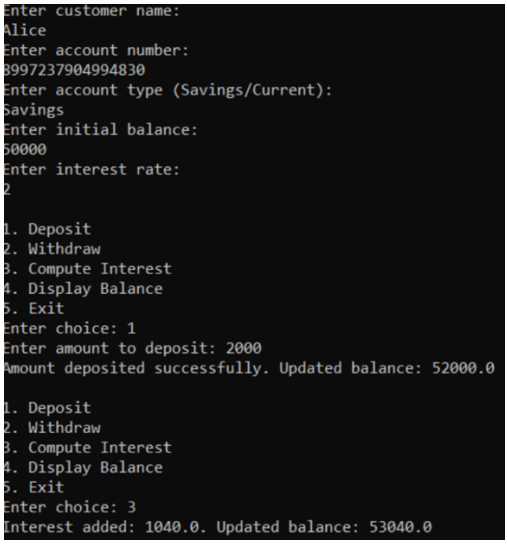
Experiment 5

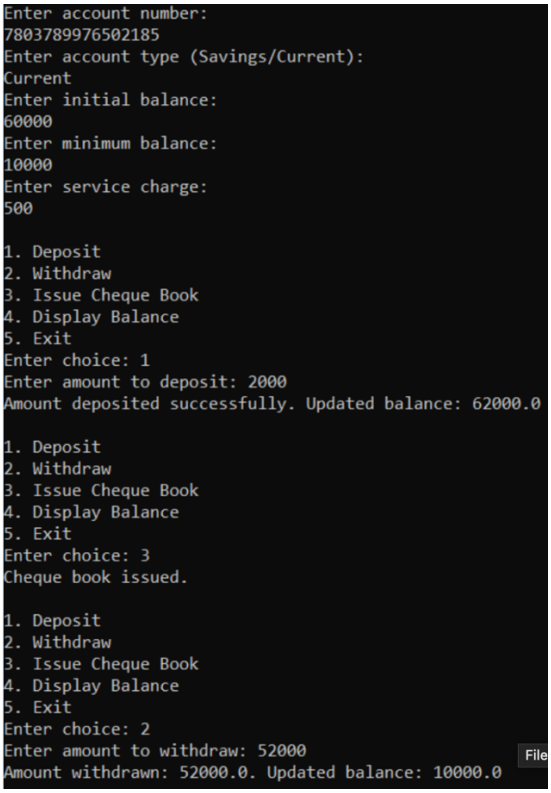
```
import java.util.Scanner;
class Account {
   protected String customerName;
   protected String accountNumber;
   protected String accountType;
   protected double balance;

   public Account(String customerName, String accountNumber, String accountType, double
initialBalance) {
      this.customerName = customerName;
      this.accountNumber = accountNumber;
      this.accountType = accountType;
      this.balance = initialBalance;
   }

   public void deposit(double amount) {
      if (amount > 0) {
         balance += amount;
         System.out.println("Amount deposited successfully.");
      } else {
         System.out.println("Invalid deposit amount.");
      }
   }

   public void displayBalance() {
      System.out.println("Account Balance: " + balance);
   }
}

class SavAcct extends Account {
   private double interestRate;

   public SavAcct(String customerName, String accountNumber, double initialBalance,
double interestRate) {
      super(customerName, accountNumber, "Savings", initialBalance);
      this.interestRate = interestRate;
   }

   public void computeAndDepositInterest() {
      double interest = balance * interestRate / 100;
      balance += interest;
      System.out.println("Interest of " + interest + " has been deposited.");
   }

   public void withdraw(double amount) {
      if (amount > 0 && amount <= balance) {
         balance -= amount;
```

```
            System.out.println("Withdrawal successful. New balance: " + balance);
        } else {
            System.out.println("Invalid withdrawal amount or insufficient balance.");
        }
    }
}

class CurAcct extends Account {
    private static final double MINIMUM_BALANCE = 500.0;
    private static final double PENALTY = 50.0;

    public CurAcct(String customerName, String accountNumber, double initialBalance) {
        super(customerName, accountNumber, "Current", initialBalance);
    }

    public void checkAndImposePenalty() {
        if (balance < MINIMUM_BALANCE) {
            balance -= PENALTY;
            System.out.println("Penalty of " + PENALTY + " imposed due to insufficient
balance.");
        }
    }

    public void withdraw(double amount) {
        if (amount > 0 && amount <= balance) {
            balance -= amount;
            System.out.println("Withdrawal successful. New balance: " + balance);
            checkAndImposePenalty();
        } else {
            System.out.println("Invalid withdrawal amount or insufficient balance.");
        }
    }
}

public class Bank {
    public static void main(String[] args) {
        SavAcct savingsAccount = new SavAcct("Alice", "SA12345", 1000.0, 5.0);
        CurAcct currentAccount = new CurAcct("Bob", "CA54321", 600.0);

        // Savings account operations
        System.out.println("Savings Account Operations:");
        savingsAccount.deposit(500);
        savingsAccount.displayBalance();
        savingsAccount.computeAndDepositInterest();
        savingsAccount.withdraw(300);
        savingsAccount.displayBalance();

        // Current account operations
        System.out.println("\nCurrent Account Operations:");
        currentAccount.deposit(200);
```

```
        currentAccount.displayBalance();
        currentAccount.withdraw(100);
        currentAccount.displayBalance();
        currentAccount.withdraw(700);
        currentAccount.displayBalance();
    }
}
```

OUTPUT:





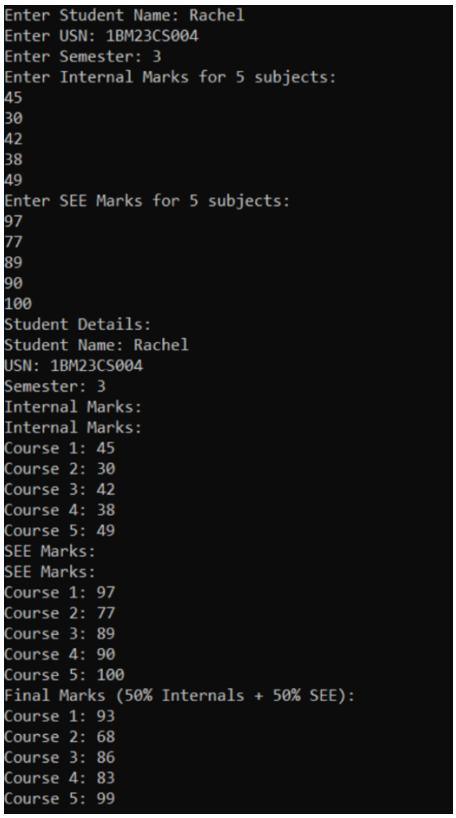
Experiment 6

```
import CIE.*;
import SEE.*;
import java.util.Scanner;

class Marks{
public static void main(String args[]){
Scanner sc=new Scanner(System.in);

System.out.println("Enter internal marks for 5 courses (out of 50): ");
int internalMarks[] = new int[5];
for (int j=0;j<5;j++)
{
System.out.print("Course " + (j + 1) + " internal marks: ");
internalMarks[j] = sc.nextInt();
}
Internals im = new Internals(internalMarks);
System.out.println("Enter SEE marks for 5 courses (out of 50): ");
int seeMarks[] = new int[5];
for (int j=0;j<5;j++)
{
System.out.print("Course "+(j + 1)+" SEE marks: ");
seeMarks[j] = sc.nextInt();
}
Externals ext=new Externals("Thanu","yy78",2,seeMarks);
System.out.println("\nFinal Marks for " + ext.name + ":" + ext.usn + ":for:");
System.out.println("Semester: " + ext.sem);
for(int x=0;x<5;x++){
int total=0;
total=im.m[x]+ext.m[x];
System.out.println("subject "+(x+1)+":  "+total);
}
}
}
```

```
package CIE;

public class Internals{
        public int[] m = new int[5];
        public Internals(int[] marks) {
        for (int i = 0; i < 5; i++) {
                m[i] = marks[i];
        }
        }
}
```

```
package CIE;

public class Student{
        public String usn;
        public String name;
        public int sem;
        public Student(String n, String u,int s){
        usn=u;
        name=n;
        sem=s;
        }
}

package SEE;
import CIE.Student;

public class Externals extends Student{
public int[] m = new int[5];
public Externals(String name, String usn, int sem,int seeMarks[]){
super(name,usn,sem);
for (int i = 0; i < 5; i++)
{
this.m[i] = seeMarks[i];
}
}
}
```

OUTPUT:



Experiment 7

```
class WrongAgeException extends Exception {
   public WrongAgeException(String message) {
      super(message);
   }
}

class Father {
   protected int age;

   public Father(int age) throws WrongAgeException {
      if (age < 0) {
         throw new WrongAgeException("Father's age cannot be negative!");
      }
      this.age = age;
   }
}

class Son extends Father {
   private int sonAge;

   public Son(int fatherAge, int sonAge) throws WrongAgeException {
      super(fatherAge);
      if (sonAge < 0) {
         throw new WrongAgeException("Son's age cannot be negative!");
      }
      if (sonAge >= fatherAge) {
         throw new WrongAgeException("Son's age cannot be greater than or equal to Father's
age!");
      }
      this.sonAge = sonAge;
   }

   public void displayAges() {
      System.out.println("Father's Age: " + age);
      System.out.println("Son's Age: " + sonAge);
   }
}

public class ExceptionHandling {
   public static void main(String[] args) {
      try {
         Father father = new Father(40);
         Son son = new Son(40, 20);
         son.displayAges();
         Father invalidFather = new Father(-5);
      } catch (WrongAgeException e) {
         System.out.println("Exception: " + e.getMessage());
```

```
        }
        try {
            Son invalidSon = new Son(30, 35);
        } catch (WrongAgeException e) {
            System.out.println("Exception: " + e.getMessage());
        }
    }
}
```

OUTPUT:

```
Enter Father's Age:
45
Enter Son's Age:
65
Father's age: 45
SonOlderThanFatherException: Son's age cannot be greater than or equal to Father's age.
```

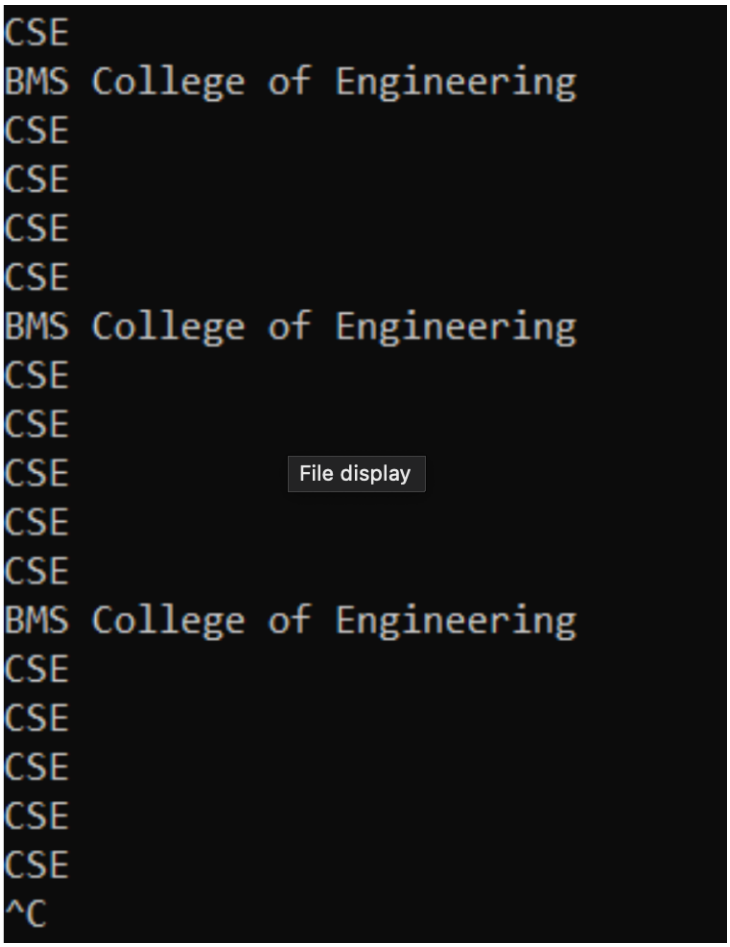
```
Enter Father's Age:
-20
Enter Son's Age:
5
WrongAgeException: Age cannot be negative.
```

```
Enter Father's Age:
55
Enter Son's Age:
20
Father's age: 55
Son's age: 20
```

Experiment 8

```
class CollegeThread extends Thread {
   public void run() {
      try {
         for (int i = 0; i < 5; i++) {
            System.out.println("BMS College of Engineering");
            Thread.sleep(10000);
         }
      } catch (InterruptedException e) {
         System.out.println("CollegeThread interrupted.");
      }
   }
}

class CSEThread extends Thread {
   public void run() {
      try {
         for (int i = 0; i < 25; i++) {
            System.out.println("CSE");
            Thread.sleep(2000);
         }
      } catch (InterruptedException e) {
         System.out.println("CSEThread interrupted.");
      }
   }
}

public class ThreadExample {
   public static void main(String[] args) {
      CollegeThread collegeThread = new CollegeThread();
      CSEThread cseThread = new CSEThread();

      collegeThread.start();
      cseThread.start();
   }
}
```

OUTPUT:

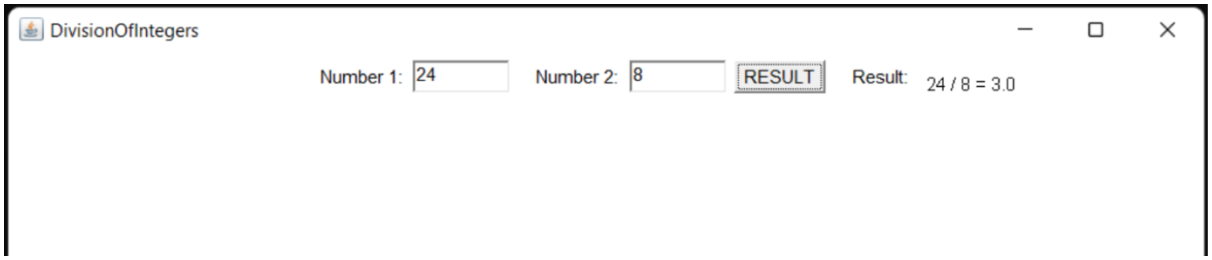

CSE
BMS College of Engineering
CSE
CSE
CSE
CSE
BMS College of Engineering
CSE
CSE
CSE
File display
CSE
CSE
BMS College of Engineering
CSE
CSE
CSE
CSE
CSE
^C

Experiment 9

```
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;

public class IntDiv extends JFrame {
   private JTextField txtNum1, txtNum2, txtResult;
   private JButton btnDivide;

   public IntDiv() {
      setTitle("Integer Division Calculator");
      setSize(400, 250);
      setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
      setLayout(new GridLayout(4, 2, 10, 10));

      initializeComponents();

      addComponentsToFrame();

      setLocationRelativeTo(null);
   }

   private void initializeComponents() {

      JLabel lblNum1 = new JLabel("Number 1:");
      JLabel lblNum2 = new JLabel("Number 2:");
      JLabel lblResult = new JLabel("Result:");

      txtNum1 = new JTextField();
      txtNum2 = new JTextField();
      txtResult = new JTextField();
      txtResult.setEditable(false);

      btnDivide = new JButton("Divide");
      btnDivide.addActionListener(new ActionListener() {
         @Override
         public void actionPerformed(ActionEvent e) {
            performDivision();
         }
      });
   }

   private void addComponentsToFrame() {
      add(new JLabel("Number 1:"));
      add(txtNum1);
      add(new JLabel("Number 2:"));
      add(txtNum2);
      add(new JLabel("Result:"));
```

```
        add(txtResult);
        add(new JLabel());
        add(btnDivide);
    }

    private void performDivision() {
        try {
            int num1 = Integer.parseInt(txtNum1.getText());
            int num2 = Integer.parseInt(txtNum2.getText());
            int result = num1 / num2;
            txtResult.setText(String.valueOf(result));

        } catch (NumberFormatException nfe) {
            JOptionPane.showMessageDialog(
                this,
                "Please enter valid integers!",
                "Input Error",
                JOptionPane.ERROR_MESSAGE
            );
        } catch (ArithmeticException ae) {
            JOptionPane.showMessageDialog(
                this,
                "Cannot divide by zero!",
                "Arithmetic Error",
                JOptionPane.ERROR_MESSAGE
            );
        }
    }

    public static void main(String[] args) {
        SwingUtilities.invokeLater(new Runnable() {
            public void run() {
                new IntDiv().setVisible(true);
            }
        });
    }


}
```

OUTPUT:



Experiment 10

```
class AA {
  synchronized void foo(BB b) {
        String name = Thread.currentThread().getName();
        System.out.println(name + " entered A.foo");
        try {
              Thread.sleep(1000);
        } catch(Exception e) {
              System.out.println("A Interrupted");
        }
        System.out.println(name + " trying to call B.last()");
        b.last();
  }
  synchronized void last() {
        System.out.println("Inside A.last");
  }
}

class BB {
  synchronized void bar(AA a) {
    String name = Thread.currentThread().getName();
        System.out.println(name + " entered BB.bar");
        try {
              Thread.sleep(1000);
        } catch(Exception e) {
              System.out.println("B Interrupted");
        }
        System.out.println(name + " trying to call A.last()");
        a.last();
   }
   synchronized void last() {
        System.out.println("Inside A.last");
   }
}

class Deadlock implements Runnable {
        AA a = new AA();
        BB b = new BB();
        Deadlock() {
              Thread.currentThread().setName("MainThread");
              Thread t = new Thread(this, "RacingThread");
              t.start();
              a.foo(b); // get lock on a in this thread.
              System.out.println("Back in main thread");
        }
        public void run() {
              b.bar(a); // get lock on b in other thread.
              System.out.println("Back in other thread");
```

```
	}
	public static void main(String args[]) {
		new Deadlock();
	}
}

//INTERPROCESS COMMUNICATION

class Q {
	int n;
	synchronized int get() {
		System.out.println("Got: " + n);
		return n;
	}
	synchronized void put(int n) {
		this.n = n;
		System.out.println("Put: " + n);
	}
}

class Producer implements Runnable {
	Q q;
	Producer(Q q) {
		this.q = q;
		new Thread(this, "Producer").start();
	}
	public void run() {
		int i = 0;
		while(true) {
			q.put(i++);
		}
	}
}

class Consumer implements Runnable {
	Q q;
	Consumer(Q q) {
		this.q = q;
		new Thread(this, "Consumer").start();
	}
	public void run() {
		while(true) {
			q.get();
		}
	}
}

class Deadlock {
	public static void main(String args[]) {
		Q q = new Q();
```

```
                new Producer(q);
                new Consumer(q);
                System.out.println("Press Control-C to stop.");
        }
}
```

OUTPUT: